

SKILL PROBLEMS

5.

```
sai_pranavi@LAPTOP-RVOHSI x + v
sai_pranavi@LAPTOP-RVOHSPKQ:~/exp5$ nano single_quotes.c
sai_pranavi@LAPTOP-RVOHSPKQ:~/exp5$ gcc single_quotes.c -o single_quotes
sai_pranavi@LAPTOP-RVOHSPKQ:~/exp5$ ./single_quotes
Enter text inside single quotes: '$HOME hello'
Literal content: '$HOME hello'
sai_pranavi@LAPTOP-RVOHSPKQ:~/exp5$ nano double_quotes.c
sai_pranavi@LAPTOP-RVOHSPKQ:~/exp5$ gcc double_quotes.c -o double_quotes
sai_pranavi@LAPTOP-RVOHSPKQ:~/exp5$ ./double_quotes
Double quoted string: "Rahul"
sai_pranavi@LAPTOP-RVOHSPKQ:~/exp5$
```

```
sai_pranavi@LAPTOP-RVOHSI x + v
GNU nano 8.7.1 single_quotes.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main() {
    char input[100];

    printf("Enter text inside single quotes: ");
    fgets(input, sizeof(input), stdin);

    printf("Literal content: %s", input);

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSI x + v
GNU nano 8.7.1 double_quotes.c
#include <stdio.h>
#include <stdlib.h>

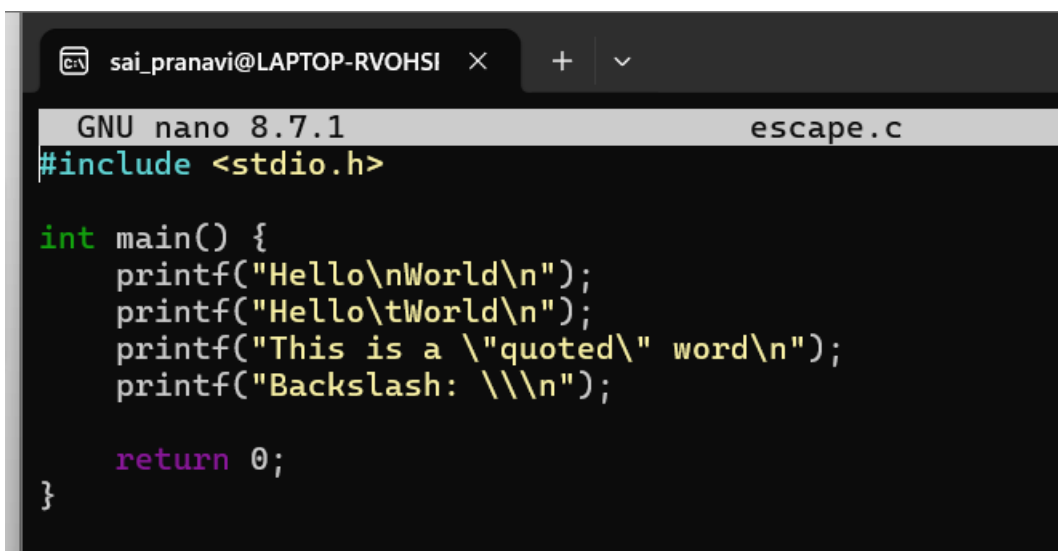
int main() {
    char *name = "Rahul";

    printf("Double quoted string: \"%s\\n\"", name);

    return 0;
}
```

6.

```
sai_pranavi@LAPTOP-RVOHSPKQ:~/exp5$ nano escape.c
sai_pranavi@LAPTOP-RVOHSPKQ:~/exp5$ gcc escape.c -o escape
sai_pranavi@LAPTOP-RVOHSPKQ:~/exp5$ ./escape
Hello
World
Hello    World
This is a "quoted" word
Backslash: \
sai_pranavi@LAPTOP-RVOHSPKQ:~/exp5$ nano child.c
sai_pranavi@LAPTOP-RVOHSPKQ:~/exp5$ gcc child.c -o child
sai_pranavi@LAPTOP-RVOHSPKQ:~/exp5$ ./child
Child process created
total 80
-rwxr-xr-x 1 sai_pranavi sai_pranavi 16080 Sep 26 05:10 child
-rw-r--r-- 1 sai_pranavi sai_pranavi  422 Sep 26 05:10 child.c
-rwxr-xr-x 1 sai_pranavi sai_pranavi 15960 Sep 26 05:06 double_quotes
-rw-r--r-- 1 sai_pranavi sai_pranavi  149 Sep 26 05:06 double_quotes.c
-rwxr-xr-x 1 sai_pranavi sai_pranavi 15952 Sep 26 05:09 escape
-rw-r--r-- 1 sai_pranavi sai_pranavi  184 Sep 26 05:09 escape.c
-rwxr-xr-x 1 sai_pranavi sai_pranavi 16096 Sep 26 05:05 single_quotes
-rw-r--r-- 1 sai_pranavi sai_pranavi  244 Sep 26 05:05 single_quotes.c
Parent process completed
```



```
GNU nano 8.7.1 escape.c
#include <stdio.h>

int main() {
    printf("Hello\nWorld\n");
    printf("Hello\tWorld\n");
    printf("This is a \"quoted\" word\n");
    printf("Backslash: \\n");

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSI x + v
GNU nano 8.7.1 child.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid < 0) {
        printf("Fork failed\n");
    }
    else if (pid == 0) {
        printf("Child process created\n");

        execlp("ls", "ls", "-l", NULL);

        printf("Execution failed\n");
    }
    else {
        wait(NULL);
        printf("Parent process completed\n");
    }

    return 0;
}
```

7.

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ cd ~
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano waitpid.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano waitpidd.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc waitpidd.c -o waitpidd
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./waitpidd
Child process is running...
Child process completed.
Parent: Child process completed.
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano path_search.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc path_search.c -o path_search
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./path_search
Executable found: /usr/bin/ls
```

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid < 0) {
        printf("Fork failed\n");
        return 1;
    }

    if (pid == 0) {
        printf("Child process is running...\n");
        sleep(2);
        printf("Child process completed.\n");
    }
    else {
        waitpid(pid, NULL, 0);
        printf("Parent: Child process completed.\n");
    }

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSI x + v
GNU nano 8.7.1 path_search.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

int main() {
    char *path;
    char *copy;
    char *dir;

    path = getenv("PATH");

    if (path == NULL) {
        printf("PATH variable not found\n");
        return 1;
    }

    copy = strdup(path);
    dir = strtok(copy, ":");

    while (dir != NULL) {
        char fullpath[500];

        snprintf(fullpath, sizeof(fullpath), "%s/%s", dir, "ls");

        if (access(fullpath, X_OK) == 0) {
            printf("Executable found: %s\n", fullpath);
            free(copy);
            return 0;
        }

        dir = strtok(NULL, ":");
    }

    printf("Command not found\n");
}
```

8.

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano variable.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc variable.c -o variable
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./variable
Enter variable name: HOME
Value: /home/sai_pranavi
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano builtins.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc builtins.c -o builtins
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./builtins
Enter command: pwd
PWD command executed
```



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GNU nano 8.7.1

variable.c

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    char variable[50];
    char *value;

    printf("Enter variable name: ");
    scanf("%49s", variable);

    value = getenv(variable);

    if (value != NULL)
        printf("Value: %s\n", value);
    else
        printf("Variable is undefined\n");

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSI × + ▾
GNU nano 8.7.1 builtins.c
#include <stdio.h>
#include <string.h>

void pwd_command() {
    printf("PWD command executed\n");
}

void cd_command() {
    printf("CD command executed\n");
}

void exit_command() {
    printf("EXIT command executed\n");
}

int main() {
    char command[50];

    printf("Enter command: ");
    scanf("%49s", command);

    if (strcmp(command, "pwd") == 0)
        pwd_command();
    else if (strcmp(command, "cd") == 0)
        cd_command();
    else if (strcmp(command, "exit") == 0)
        exit_command();
    else
        printf("Invalid command\n");

    return 0;
}
```

9.

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano changedir.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc changedir.c -o changedir
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./changedir
Enter directory path: /tmp
Directory changed successfully
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano builtins2.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc builtins2.c -o builtins2
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./builtins2
Enter command: pwd
Executing PWD
```



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GNU nano 8.7.1

changedir.c

```
#include <stdio.h>
#include <unistd.h>
#include <limits.h>

int main() {
    char path[PATH_MAX];

    printf("Enter directory path: ");
    scanf("%s", path);

    if (chdir(path) == 0)
        printf("Directory changed successfully\n");
    else
        perror("chdir");

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSI x + v
GNU nano 8.7.1 builtins2.c
#include <stdio.h>
#include <string.h>

void execute_command(char *command) {
    if (strcmp(command, "pwd") == 0) {
        printf("Executing PWD\n");
    }
    else if (strcmp(command, "cd") == 0) {
        printf("Executing CD\n");
    }
    else if (strcmp(command, "exit") == 0) {
        printf("Executing EXIT\n");
    }
    else {
        printf("Invalid built-in command\n");
    }
}

int main() {
    char command[50];

    printf("Enter command: ");
    scanf("%49s", command);

    execute_command(command);

    return 0;
}
```

10.

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano currentdir.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc currentdir.c -o currentdir
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./currentdir
Current Directory: /home/sai_pranavi
Program exiting...
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano export.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc export.c -o export
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./export
Enter variable name: MYNAME
Enter value: Pranavi
Environment variable exported successfully
MYNAME = Pranavi
```



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GNU nano 8.7.1

currentdir.c

```
#include <stdio.h>
#include <unistd.h>
#include <limits.h>

int main() {
    char path[PATH_MAX];

    if (getcwd(path, sizeof(path)) != NULL) {
        printf("Current Directory: %s\n", path);
    }
    else {
        perror("getcwd");
        return 1;
    }

    printf("Program exiting...\n");

    return 0;
}
```

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    char name[50];
    char value[100];

    printf("Enter variable name: ");
    scanf("%49s", name);

    printf("Enter value: ");
    scanf("%99s", value);

    if (setenv(name, value, 1) == 0) {
        printf("Environment variable exported successfully\n");
        printf("%s = %s\n", name, getenv(name));
    }
    else {
        perror("setenv");
    }

    return 0;
}
```